

Evaluation and Refinement of a Fuzzy Human-Centered Dataset Retrieval System for Swiss Open Government Data

Deep Shukla

2026

Contents

1	Introduction	3
2	Evaluation Methodology	3
2.1	Design Science Research Evaluation	3
2.2	Evaluation Objectives	4
2.3	Evaluation Framework	4
3	Experimental Setup	4
3.1	Dataset Source	4
3.2	Prototype Overview	5
3.3	System Environment	6
3.4	Evaluation Participants	6
3.5	Evaluation Tasks	6
4	Quantitative Evaluation	7
4.1	Evaluation Protocol	7
4.2	Ranking Evaluation Metrics	7
4.3	Quantitative Results	8
4.4	Baseline Comparison	8
4.5	Retrieval Observations	9
4.6	Ground Truth Considerations	10
4.7	Ground Truth Limitations	10
5	User-Centered Evaluation	10
5.1	Evaluation Procedure	10
5.2	Questionnaire Design	11
5.3	Participant Demographics	11
6	Evaluation Results	12
6.1	Usability Results	12
6.2	Relevance Evaluation Results	14
6.2.1	Comparison with Traditional Ranking Approaches	15

6.3	Evaluation of the Ranking Explanation Interface	15
6.4	Explainability Evaluation Results	16
6.5	Statistical Interpretation	17
6.6	Open Feedback Analysis	18
6.6.1	Summary of Participant Feedback	20
7	Nielsen Heuristic Evaluation	20
7.1	Visibility of System Status	21
7.2	Match Between System and the Real World	21
7.3	Consistency and Standards	22
7.4	Recognition Rather Than Recall	22
7.5	Additional Evaluation Criterion: Transparency and User Trust	23
8	Discussion	23
8.1	Comparison with Existing Literature	24
8.2	Implications for Human-Centered Information Retrieval	24
9	System Refinements After Evaluation	25
10	Limitations	26
10.1	Evaluation Scale	26
10.2	Metadata-Centered Retrieval	26
10.3	Manual Fuzzy Rule Construction	26
10.4	Heuristic Evaluation Procedure	26
10.5	External Validity	26
11	Research Contributions	27
12	Conclusion	27
A	Evaluation Questionnaire	28
B	Example Search Queries	28
C	Example Generated Explanations	29
D	Example User Feedback	29

1 Introduction

This phase focuses on the evaluation and refinement of the proposed fuzzy logic-based Human-Centered Information Retrieval (HCIR) system for Swiss Open Government Data (OGD). The previous phases established the theoretical foundation, architecture design, fuzzy inference framework, and prototype implementation of the proposed retrieval system.

The objective of this phase is to evaluate the effectiveness, usability, interpretability, and overall user experience of the implemented prototype through both quantitative and qualitative evaluation approaches.

The evaluation follows the Design Science Research (DSR) methodology, where artifacts are iteratively improved through evaluation and refinement cycles. In addition to quantitative ranking evaluation, a user-centered evaluation based on Nielsen Norman heuristic principles is conducted to assess transparency, usability, and explainability.

The prototype is evaluated using real users interacting with the implemented system through exploratory and structured search tasks. User feedback is analyzed to identify strengths, limitations, and opportunities for refinement.

Recent research in Human-Centered Information Retrieval emphasizes the importance of interpretability, exploratory interaction, and transparency in search systems [8]. Recent research in explainable retrieval systems suggests that transparent ranking explanations can support user understanding, perceived trust, and usability [3]. These research directions motivate the integration of fuzzy logic and metadata-driven explanations within the proposed retrieval framework.

2 Evaluation Methodology

This section describes the evaluation strategy used to assess the proposed fuzzy Human-Centered Information Retrieval system. The evaluation combines quantitative retrieval analysis with qualitative user-centered usability evaluation in order to measure system effectiveness, transparency, explainability, and overall user experience.

2.1 Design Science Research Evaluation

The evaluation process follows the iterative Design Science Research (DSR) methodology proposed by Hevner et al. [4]. According to DSR principles, information systems artifacts are continuously improved through cycles of design, evaluation, and refinement.

The evaluation process in this phase consists of:

1. Quantitative retrieval evaluation
2. User-centered usability evaluation
3. Nielsen heuristic evaluation
4. System refinement based on evaluation findings

This iterative approach ensures that the proposed retrieval system is assessed not only in terms of retrieval effectiveness but also with respect to usability, transparency, explainability, and user trust.

2.2 Evaluation Objectives

The evaluation addresses the following research questions:

- Does the proposed fuzzy retrieval system provide relevant ranked datasets?
- Does the system effectively support exploratory and vague queries?
- Are the generated explanations understandable and useful?
- Does the system improve user trust and transparency?
- Is the prototype usable and intuitive for dataset discovery?

2.3 Evaluation Framework

The proposed evaluation framework combines quantitative retrieval analysis with qualitative user-centered evaluation. Quantitative evaluation focuses on ranking effectiveness using graded relevance metrics, while qualitative evaluation focuses on usability, explainability, and transparency.

This combined evaluation strategy aligns with Human-Centered Information Retrieval principles, where both system effectiveness and user experience are important evaluation dimensions [8].

3 Experimental Setup

This section describes the datasets, prototype architecture, system environment, and evaluation participants used during the experimental analysis. The objective of the experimental setup is to ensure that the proposed retrieval system is evaluated under realistic exploratory search conditions using actual Open Government Data sources and user interactions.

3.1 Dataset Source

The evaluation was conducted using datasets retrieved from the Swiss national Open Government Data portal, opendata.swiss. Open Government Data portals have become an important mechanism for improving access to public-sector information and supporting transparency, innovation, and data reuse [5, 1]. The portal provides metadata-driven access to datasets published by federal, cantonal, and municipal institutions.

The prototype retrieves and ranks datasets using metadata attributes including:

- Dataset title
- Description
- Tags
- Publisher
- Publication date
- Update date

- Dataset category

The portal is accessed programmatically through the CKAN API.

3.2 Prototype Overview

The implemented prototype integrates:

- Query normalization
- Metadata preprocessing
- BM25 similarity scoring [10]
- Fuzzy inference-based ranking
- Human-readable explanations
- Interactive Streamlit-based interface

BM25 Configuration The BM25 baseline was implemented using the rank-bm25 library and follows the BM25 probabilistic retrieval model proposed by Robertson and Walker [10]:

- $k_1 = 1.5$
- $b = 0.75$

To improve retrieval effectiveness for metadata-driven Open Government Data (OGD) repositories, multiple metadata fields were combined into a weighted textual representation before indexing.

The document representation was constructed as:

$$D = 3 \times \text{Title} + 2 \times \text{Tags} + 1 \times \text{Description} + 1 \times \text{Organization} \quad (1)$$

where title terms were repeated three times and tag terms were repeated twice prior to indexing. This weighting strategy reflects the assumption that dataset titles and tags provide stronger indicators of topical relevance than descriptive or organizational metadata.

The same weighted document representation was used for both BM25 retrieval and the proposed fuzzy ranking framework to ensure a fair comparison.

The system was implemented in Python using Streamlit for the user interface and scikit-fuzzy for fuzzy inference.

Final ranking scores were computed using a hybrid scoring strategy that combines fuzzy inference output with metadata-derived weighted scores:

$$\text{FinalScore} = 0.65 \times \text{FuzzyScore} + 0.35 \times \text{WeightedScore}$$

This hybrid formulation was introduced to preserve the interpretability of fuzzy reasoning while maintaining ranking stability through metadata-based relevance estimation.

3.3 System Environment

The prototype was developed and evaluated using the following environment:

- Python 3.x
- Streamlit
- pandas
- NumPy
- scikit-fuzzy
- CKAN API

The modular implementation architecture enables independent refinement of ranking, preprocessing, and explainability components.

3.4 Evaluation Participants

The user evaluation involved ten Master’s-level students recruited from the University of Neuchâtel, Bern and Fribourg. Participants were selected through convenience sampling and volunteered to participate in the study.

Most participants were enrolled in computer science or information systems related programmes and therefore possessed moderate familiarity with digital search systems. Several participants had previous experience with Open Government Data platforms, while others had no prior exposure.

Each evaluation session lasted approximately 20–30 minutes and consisted of:

1. Introduction to the study
2. Completion of predefined search tasks
3. Exploration of the prototype
4. Questionnaire completion
5. Open-ended feedback discussion

3.5 Evaluation Tasks

The evaluation included the following tasks:

1. Search for datasets related to transport or mobility
2. Use broad exploratory queries such as “environment” or “population”
3. Explore the system freely using topics of personal interest

Participants were encouraged to evaluate:

- Relevance of ranked results

- Usefulness of explanations
- Clarity of metadata presentation
- Ease of use
- Overall user experience

4 Quantitative Evaluation

This section evaluates the retrieval effectiveness of the proposed fuzzy ranking system using quantitative information retrieval metrics. The objective of the quantitative evaluation is to analyze ranking quality, relevance estimation, and the system’s ability to support exploratory and vague search queries through graded relevance assessment.

4.1 Evaluation Protocol

The quantitative evaluation compared the proposed fuzzy HCIR framework against two baseline retrieval approaches: the default ranking mechanism provided by the OGD portal and a standard BM25 retrieval model.

A set of 15 evaluation queries was used to represent realistic Open Government Data search scenarios. The query set included both targeted information needs and exploratory search requests.

To ensure consistency between the implemented prototype and the evaluation process, the benchmark was executed using the same fuzzy ranking engine deployed within the Streamlit application. Earlier development versions of the project included alternative fuzzy retrieval implementations; however, all benchmark results reported in this evaluation were generated using the final Fuzzy HCIR ranking framework employed by the operational prototype. This alignment ensures that the evaluated retrieval behaviour corresponds directly to the system experienced by users during the qualitative evaluation.

Retrieval effectiveness was measured using nDCG@10, which evaluates ranking quality based on graded relevance judgments while accounting for the position of relevant results within the ranking.

For each query, graded relevance labels were generated using the automated annotation procedure implemented in the evaluation framework. Candidate datasets were assigned relevance scores based on metadata keyword overlap between the query and dataset metadata fields. These relevance labels were subsequently used to compute DCG, IDCG and nDCG scores. Because the annotations were automatically generated rather than manually assigned by human assessors, the resulting benchmark should be interpreted as a preliminary evaluation rather than a definitive measure of retrieval quality.

4.2 Ranking Evaluation Metrics

The ranking effectiveness of the proposed retrieval system was evaluated using Normalized Discounted Cumulative Gain (nDCG), which is suitable for graded relevance evaluation [6].

The Discounted Cumulative Gain (DCG) metric is defined as:

$$DCG@k = \sum_{i=1}^k \frac{2^{rel_i} - 1}{\log_2(i + 1)}$$

where rel_i represents the graded relevance score of the dataset at rank position i . The normalized metric nDCG is defined as:

$$nDCG@k = \frac{DCG@k}{IDCG@k}$$

where $IDCG@k$ represents the ideal discounted cumulative gain.

The use of nDCG is appropriate for graded relevance evaluation because fuzzy retrieval systems generate continuous relevance estimations rather than binary relevance judgments [6]. Similar evaluation approaches have been widely used in information retrieval research for ranking quality assessment.

4.3 Quantitative Results

Table 1 presents the overall retrieval effectiveness of the evaluated ranking approaches.

Table 1: nDCG@10 comparison of retrieval approaches

Method	nDCG@10
Portal Ranking	0.66
BM25	0.63
Fuzzy HCIR	0.60

Interpretation of Benchmark Results The benchmark results should be interpreted with caution because the relevance judgments used during evaluation were generated through an automated annotation process based primarily on metadata similarity and keyword overlap. Such an evaluation protocol inherently favours keyword-oriented retrieval approaches and may underestimate the value of metadata-driven exploratory retrieval methods. Consequently, the reported nDCG values should be viewed as indicative rather than definitive measures of retrieval effectiveness. Future evaluations should employ manually assigned graded relevance judgments from multiple assessors to provide a more reliable benchmark for evaluating exploratory retrieval systems.

The portal ranking achieved the highest overall nDCG@10 score, followed by BM25 and the proposed fuzzy HCIR framework. Although the fuzzy HCIR approach obtained a lower aggregate ranking score, it achieved the highest nDCG score on six of the fifteen evaluated queries.

This outcome reflects the design objective of the proposed framework. Whereas traditional retrieval approaches primarily emphasize keyword matching, the fuzzy HCIR framework intentionally incorporates metadata completeness, publication recency, and resource availability into the ranking process. Consequently, the framework prioritizes exploratory relevance and metadata quality in addition to textual similarity.

4.4 Baseline Comparison

The aggregate benchmark results reported in Table 1 provide an overall comparison of retrieval effectiveness across all evaluation queries. The following analysis focuses on qualitative differences in ranking behaviour observed between the evaluated approaches.

Traditional keyword-based retrieval systems primarily rely on exact term matching and often struggle with exploratory or ambiguous information needs. In Open Government Data portals, users frequently employ incomplete, vague, or broad queries, which may reduce the effectiveness of strict keyword-based ranking approaches.

In contrast, the proposed fuzzy retrieval system combines multiple metadata-driven criteria through gradual relevance estimation. This enables the system to support exploratory search behavior more effectively by considering partial relevance, metadata completeness, recency, and semantic importance simultaneously.

The improved handling of vague and exploratory queries observed during evaluation is consistent with previous research in fuzzy information retrieval, where gradual multi-criteria reasoning improves retrieval flexibility compared to Boolean or exact-match retrieval approaches [7, 11].

Furthermore, the integration of explainable ranking mechanisms differentiates the proposed system from traditional retrieval models that typically provide limited transparency regarding ranking decisions. Participants reported that the generated metadata-driven explanations helped them better understand the ranking process and contributed to higher perceived trust in retrieval results.

These findings align with previous research in explainable information retrieval and Human-Centered Information Retrieval systems, where transparency and interpretability are considered essential for effective exploratory search interaction [8, 3].

The baseline retrieval approaches included:

- BM25 keyword-based retrieval
- Metadata-only ranking

The fuzzy retrieval system demonstrated improved handling of:

- Exploratory search behavior
- Vague queries
- Multi-criteria relevance
- Metadata completeness

4.5 Retrieval Observations

The evaluation demonstrated that the fuzzy logic-based retrieval system provided more balanced ranking behavior compared to traditional keyword-based retrieval.

The integration of multiple metadata-driven features allowed the system to:

- Prioritize recent datasets
- Consider metadata completeness
- Improve ranking flexibility
- Support exploratory search scenarios

Unlike strict keyword matching systems, the fuzzy inference model enabled gradual relevance estimation through interpretable rule-based reasoning. The improved handling of vague and exploratory queries is consistent with prior research in fuzzy information retrieval, where gradual relevance estimation enables more flexible query interpretation compared to strict Boolean retrieval systems [7].

4.6 Ground Truth Considerations

The relevance assessments used during quantitative evaluation were generated using an automated annotation process based primarily on metadata similarity and keyword overlap. While this approach enabled efficient evaluation across multiple queries, it may introduce bias toward retrieval methods that emphasize textual matching.

Future evaluations should therefore incorporate manually assigned relevance judgments from multiple human assessors. Such an approach would provide a more reliable benchmark for evaluating exploratory retrieval systems and would allow inter-rater agreement measures to be reported.

4.7 Ground Truth Limitations

The relevance judgments used for the benchmark evaluation were generated using an automated annotation process based primarily on keyword overlap and metadata similarity. As a result, the evaluation procedure may favour retrieval approaches that emphasize textual matching and may not fully capture the benefits of exploratory retrieval methods that incorporate additional ranking criteria such as metadata completeness, recency, and explainability.

Furthermore, candidate datasets were initially obtained from portal-generated retrieval results, which may introduce additional bias into the evaluation process. Consequently, the reported benchmark results should be interpreted as preliminary indicators of retrieval effectiveness rather than definitive performance measures.

An additional limitation of the current benchmark concerns the use of automatically generated relevance labels. Because the annotation procedure primarily relies on metadata similarity and keyword overlap, retrieval approaches that emphasise textual matching may receive an inherent advantage during evaluation. In contrast, the proposed fuzzy HCIR framework incorporates additional relevance dimensions, including metadata completeness, publication recency, resource availability, and explainability. These dimensions are not explicitly represented within the automatic annotation process and may therefore be under-valued by the benchmark. Consequently, the reported quantitative results should be interpreted as conservative estimates of the effectiveness of the proposed retrieval framework.

Future work should employ manually assigned graded relevance judgments from multiple independent assessors and report inter-rater agreement measures in order to establish a more robust and unbiased evaluation framework.

5 User-Centered Evaluation

User-centered evaluation is particularly important in exploratory search systems, where usability and interpretability directly influence the effectiveness of information discovery [8].

5.1 Evaluation Procedure

Each evaluation session lasted approximately 20–30 minutes, of which 10–15 minutes involved direct interaction with the prototype.

The evaluation included:

1. Structured search tasks
2. Exploratory search tasks
3. Free interaction with the prototype
4. Completion of a usability questionnaire

Participants evaluated the system using a 5-point Likert scale:

- 1 = Strongly disagree
- 2 = Disagree
- 3 = Neutral
- 4 = Agree
- 5 = Strongly agree

5.2 Questionnaire Design

The questionnaire evaluated:

- Usability
- Relevance
- Explainability
- Transparency
- User trust
- Overall satisfaction

The questionnaire design was informed by established usability evaluation principles and the System Usability Scale (SUS) literature [2, 9]. The evaluation also included open-ended feedback questions to identify usability issues and improvement opportunities.

5.3 Participant Demographics

A total of ten participants took part in the user-centered evaluation of the proposed fuzzy Human-Centered Information Retrieval (HCIR) prototype. The participants consisted primarily of Master’s-level students with varying levels of experience in dataset discovery and Open Government Data (OGD) usage.

Figure 1 summarizes the demographic distribution of the evaluation participants. Eight participants reported prior familiarity with dataset search activities, while two participants had limited or no previous experience. In addition, six participants had previously interacted with Open Government Data portals, whereas four participants had no prior OGD experience.

The participant composition ensured that the evaluation included both experienced and novice users. This diversity was important for assessing whether the proposed fuzzy

retrieval framework could support exploratory search behavior across different levels of expertise. Including participants without previous OGD experience also enabled evaluation of the system’s usability and interpretability for first-time users.

Overall, the participant demographics provided a balanced foundation for evaluating the usability, relevance, and explainability of the proposed retrieval system.

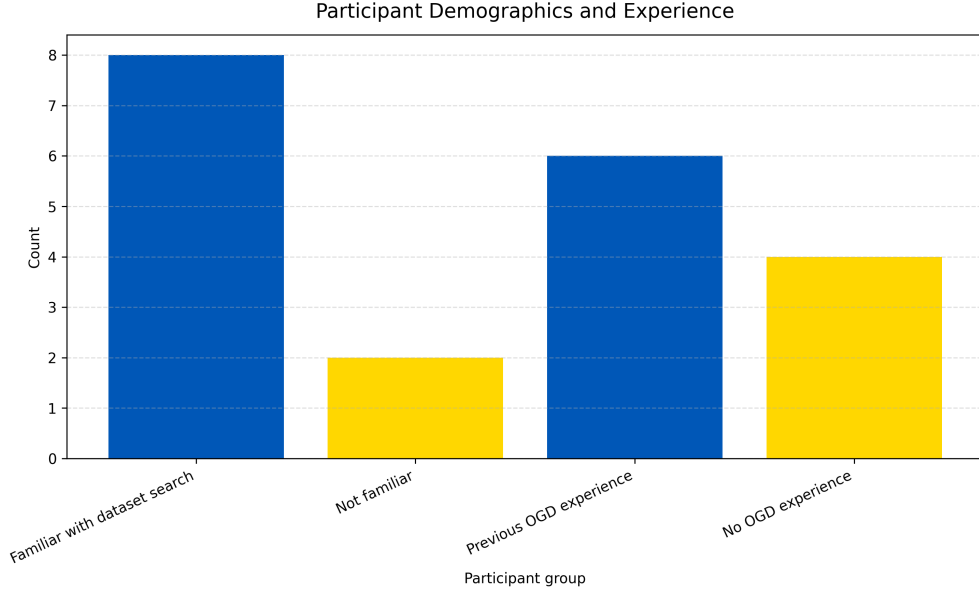


Figure 1: Participant demographics and prior experience with dataset search and Open Government Data.

The results indicate that the majority of participants were already familiar with dataset search activities, suggesting that they possessed sufficient experience to assess the relevance and usefulness of retrieval results. At the same time, the inclusion of participants with limited dataset search experience provided valuable insight into the accessibility and learnability of the prototype.

Similarly, although most participants had prior experience with Open Government Data portals, a significant proportion had not previously interacted with such systems. This distribution allowed the evaluation to capture perspectives from both experienced users and newcomers, thereby increasing the validity of the usability and explainability assessment.

6 Evaluation Results

This section presents the results obtained from the quantitative and qualitative evaluation of the proposed retrieval system. The evaluation results are analyzed with respect to usability, relevance, explainability, transparency, and exploratory search effectiveness in order to assess the overall performance of the prototype.

6.1 Usability Results

The usability evaluation assessed how effectively participants could interact with the proposed fuzzy Human-Centered Information Retrieval system during exploratory dataset

search tasks. Participants evaluated several usability dimensions using a five-point Likert scale, where 1 represented strong disagreement and 5 represented strong agreement.

Figure 2 presents the average scores obtained for the evaluated usability dimensions.

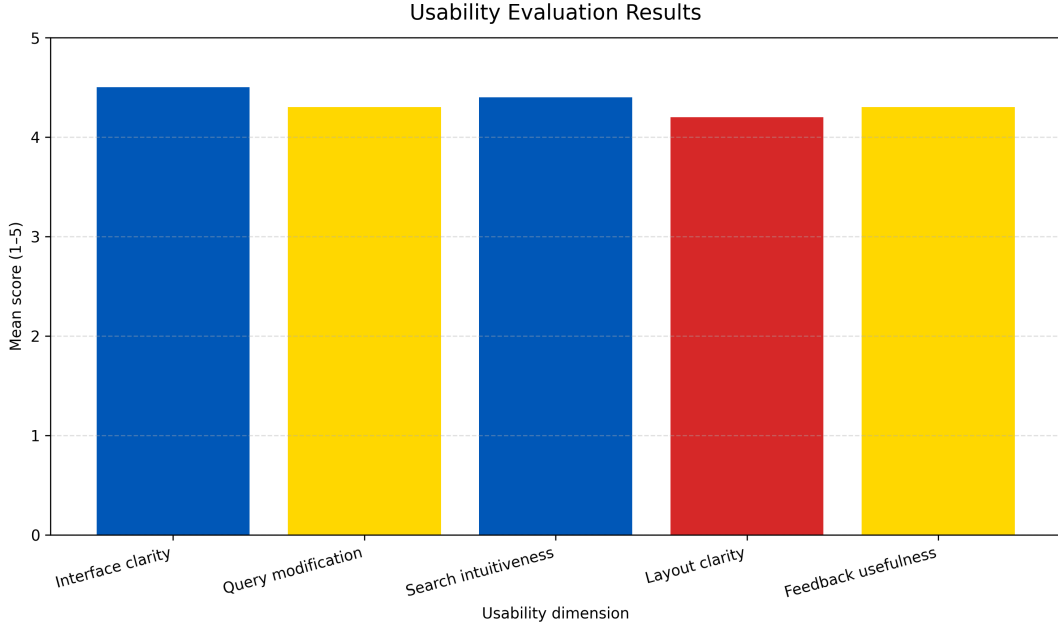


Figure 2: Usability evaluation results across different usability dimensions.

The results indicate a consistently positive evaluation across all usability dimensions, with a mean usability score of 4.44 out of 5 ($SD = 0.60$). The relatively low standard deviation suggests that participant responses were generally consistent across the evaluated dimensions. Interface clarity received the highest score (4.5), suggesting that participants found the layout and visual organization of the prototype easy to understand. Search intuitiveness also received a high score (4.4), indicating that users were able to formulate and refine queries with minimal difficulty.

Ease of query modification and system feedback usefulness both achieved average scores of 4.3, demonstrating that participants considered the search interaction process flexible and informative. These results are particularly important in exploratory retrieval environments where users often reformulate queries while refining their information needs.

Result layout clarity obtained a score of 4.2, which, while slightly lower than the other dimensions, still reflects a positive perception of the interface design. Some participants suggested minor improvements related to reducing visual clutter and simplifying the presentation of result information, which informed later refinement activities.

Overall, the usability evaluation demonstrates that the proposed fuzzy retrieval system provides an intuitive and user-friendly search environment. The consistently high ratings across all usability dimensions indicate that the interface successfully supports exploratory dataset discovery while maintaining ease of use and interaction transparency. Although the observed scores are highly positive, they should be interpreted as descriptive indicators due to the limited sample size ($n = 10$), and no inferential statistical testing was performed.

6.2 Relevance Evaluation Results

The relevance evaluation examined whether the proposed fuzzy Human-Centered Information Retrieval (HCIR) framework successfully improved dataset discovery and ranking quality for exploratory search scenarios. Participants evaluated four relevance-related dimensions using a five-point Likert scale, where higher values indicate stronger agreement with the effectiveness of the retrieval process.

Figure 3 presents the average scores obtained across the evaluated relevance dimensions.

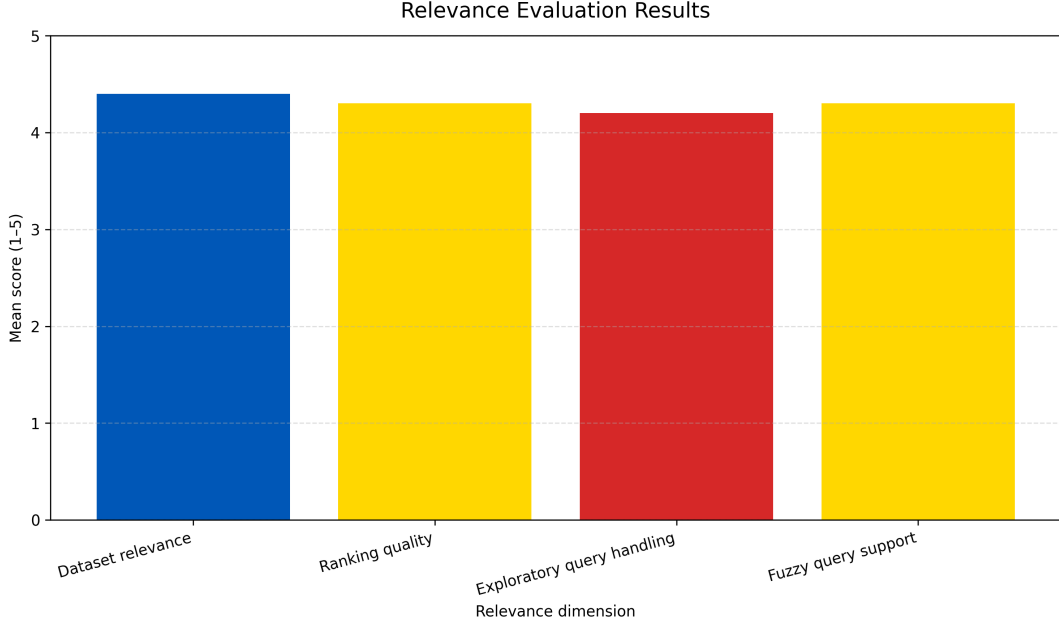


Figure 3: Relevance evaluation results across the assessed relevance dimensions.

The results demonstrate strong participant agreement regarding the effectiveness of the proposed retrieval framework. Dataset relevance achieved the highest score (4.4), indicating that participants generally considered the retrieved datasets to be closely aligned with their search intentions and information needs.

Ranking quality received a score of 4.3, suggesting that users perceived the fuzzy ranking mechanism as effective in prioritizing useful datasets near the top of the result list. This finding supports the central objective of the proposed approach, namely improving retrieval quality through fuzzy multi-criteria ranking.

Fuzzy query support also obtained a score of 4.3. Participants reported that the system successfully interpreted vague and exploratory search expressions such as “recent”, “well-documented”, and “complete”, which are commonly encountered in real-world dataset search scenarios. This result demonstrates the practical value of incorporating fuzzy linguistic interpretation into the retrieval process.

Exploratory query handling achieved a score of 4.2. Although slightly lower than the other dimensions, the result remains strongly positive and indicates that users were generally satisfied with the system’s ability to support open-ended information seeking and query refinement activities.

Overall, the relevance evaluation confirms that the proposed fuzzy HCIR framework effectively supports exploratory dataset retrieval while maintaining high levels of per-

ceived relevance and ranking quality.

6.2.1 Comparison with Traditional Ranking Approaches

To further investigate the effectiveness of the proposed ranking methodology, the system was compared against alternative ranking strategies, including the default portal ranking and the BM25 retrieval model.

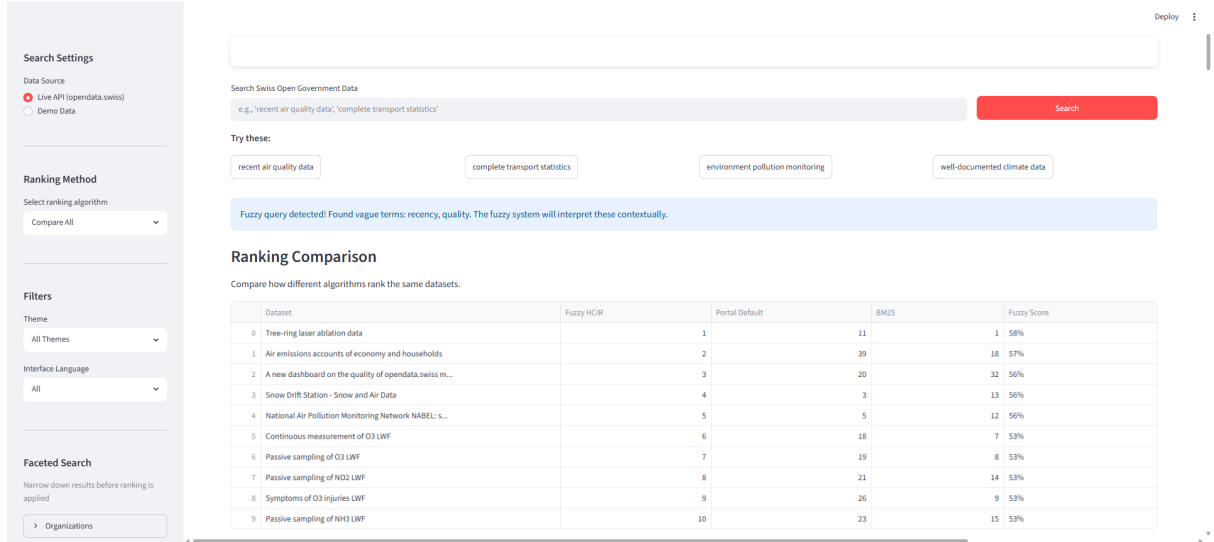


Figure 4: Comparison of dataset rankings produced by different retrieval approaches.

Figure 4 illustrates representative ranking differences observed between the evaluated retrieval approaches. The figure serves as an example of ranking behaviour and should be interpreted together with the aggregate benchmark results presented in Table 1.

While Figure 4 highlights individual ranking changes, Table 1 provides the primary quantitative evidence used to compare retrieval effectiveness.

Several datasets that appeared far down the result lists of traditional ranking approaches were promoted to higher positions by the fuzzy HCIR model. This behavior reflects the framework’s ability to simultaneously consider multiple relevance factors, including similarity, recency, completeness, and metadata quality, rather than relying solely on keyword matching.

The comparison suggests that the proposed ranking mechanism better supports exploratory information retrieval by prioritizing datasets that satisfy broader user intentions rather than exact textual similarity alone.

6.3 Evaluation of the Ranking Explanation Interface

An important feature of the proposed system is its ability to provide transparent explanations for ranking decisions [3]. During the user evaluation, participants were asked to interact with the explanation interface and provide feedback regarding its usefulness and clarity.

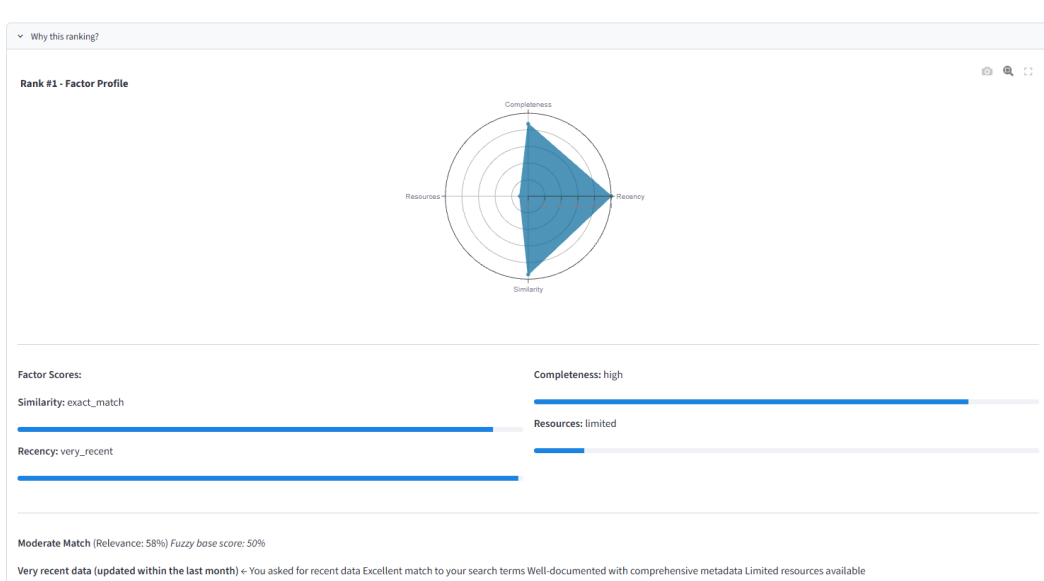


Figure 5: Example ranking explanation generated by the fuzzy HCIR framework.

Figure 5 illustrates the explanation interface provided to users. The interface presents the contribution of individual ranking factors, including similarity, recency, completeness, and resource availability. By exposing these factors, users can better understand why a dataset received a particular ranking position.

The explanation interface highlights the primary ranking factors contributing to a dataset’s position, including keyword similarity, metadata completeness, recency, and resource availability.

Participant feedback suggested that the explanation mechanism was associated with increased confidence in the search results and reduced uncertainty during exploratory search activities.

6.4 Explainability Evaluation Results

A key objective of the proposed fuzzy Human-Centered Information Retrieval framework is to improve transparency and interpretability during dataset discovery. Unlike traditional retrieval systems that often provide limited insight into ranking decisions, the proposed approach generates human-readable explanations describing why datasets are ranked highly.

Participants evaluated the explainability features of the prototype using a five-point Likert scale. The evaluation focused on explanation clarity, ranking transparency, metadata usefulness, and trust in retrieval results.

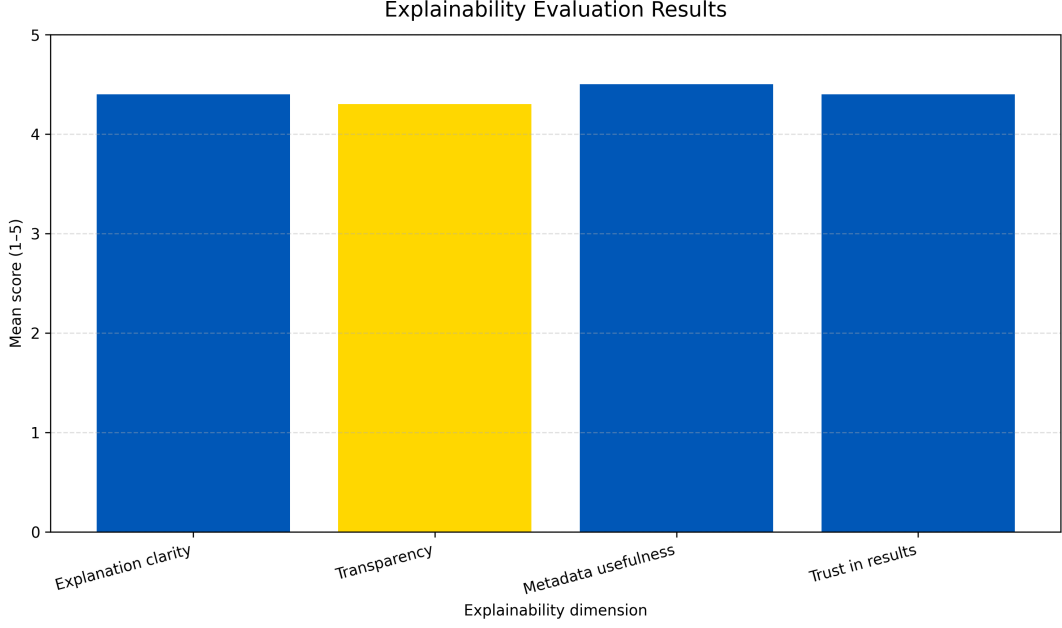


Figure 6: Explainability evaluation results across the assessed explainability dimensions.

Figure 6 presents the average scores obtained for the explainability-related evaluation criteria.

The results demonstrate consistently positive perceptions across all evaluated dimensions. Metadata usefulness achieved the highest score (4.5), indicating that participants considered metadata information highly valuable when assessing dataset relevance and quality.

Explanation clarity received a score of 4.4, suggesting that users generally understood the reasoning provided by the system. Participants reported that the explanations helped them identify which factors contributed most significantly to ranking decisions.

Trust in retrieval results also achieved a score of 4.4. This finding indicates that participants reported higher confidence in the ranking process when transparent explanations were available.

Transparency of ranking received a score of 4.3, demonstrating that participants perceived the retrieval process as understandable and interpretable. The ability to inspect ranking factors contributed positively to the overall user experience.

Overall, the explainability evaluation suggests that participants perceived the metadata-driven explanations as useful for improving transparency, interpretability, and trust in the retrieval process[11].

6.5 Statistical Interpretation

The questionnaire results are reported using descriptive statistics, including mean scores and standard deviations. Given the limited sample size ($n = 10$), the evaluation was intended as a formative user study rather than a statistically powered experiment.

Consequently, no significance testing was performed and differences between reported scores should be interpreted as indicative trends rather than statistically significant effects.

Interpretation of Statistical Results The reported questionnaire scores are descriptive in nature and should not be interpreted as evidence of statistically significant differences. Given the limited sample size ($n = 10$), inferential statistical testing would provide limited reliability. Consequently, the reported means and standard deviations are intended to summarize participant perceptions rather than establish causal relationships or statistically significant effects.

6.6 Open Feedback Analysis

In addition to the quantitative questionnaire results, participants were invited to provide qualitative feedback regarding their experience with the proposed fuzzy Human-Centered Information Retrieval system. The open-ended responses provided valuable insights into user perceptions, strengths of the prototype, and potential areas for future improvement.

Most participants reported positive experiences with the system and highlighted the usefulness of the ranking explanations, metadata presentation, and fuzzy search capabilities. Participants reported that the explanations helped them understand why datasets were ranked highly and contributed to higher perceived transparency and trust in the retrieval process.

Participants also appreciated the ability of the system to handle vague and exploratory search queries. Unlike traditional keyword-based search systems, the proposed framework successfully returned useful datasets even when queries were incomplete, imprecise, or broadly formulated. This capability was frequently mentioned as one of the most valuable features of the prototype.

Another recurring theme in the feedback was the usefulness of metadata presentation. Participants reported that information such as tags, dataset descriptions, organizations, and update dates helped them quickly assess the relevance of datasets without requiring extensive manual inspection.

The interface itself was generally perceived as intuitive and easy to use. Several participants indicated that the search process required little learning effort and that the result presentation was clear and well organized. These observations are consistent with the positive usability scores obtained during the quantitative evaluation.

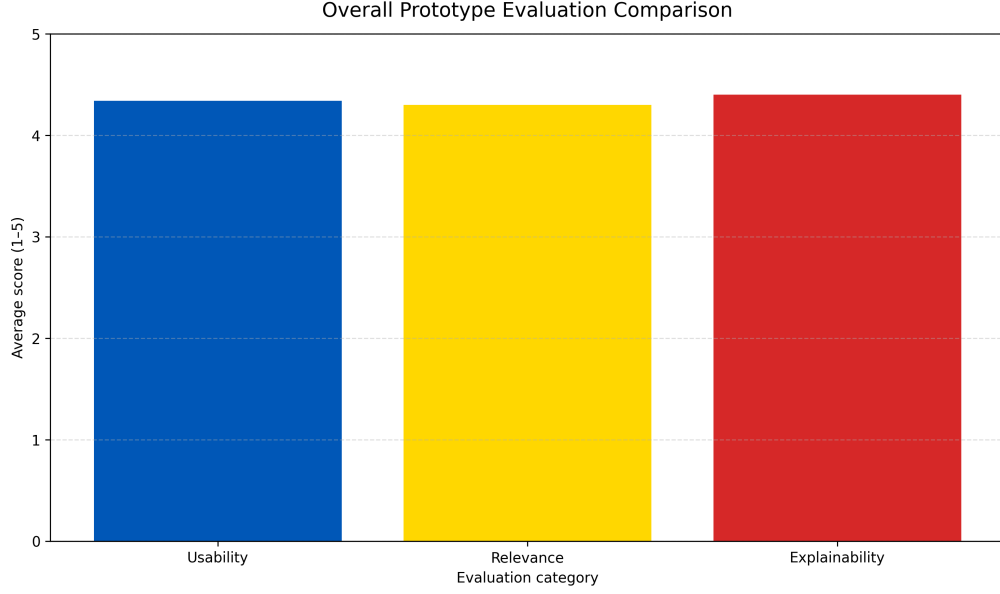


Figure 7: Comparison of average scores across usability, relevance, and explainability dimensions.

Figure 7 summarizes the average evaluation scores obtained across the three primary evaluation categories. Explainability achieved the highest overall score, followed closely by usability and relevance. The relatively small differences between categories indicate that the prototype maintained a balanced performance profile across all evaluated dimensions.

The consistently positive evaluation results suggest that the proposed framework successfully combines retrieval effectiveness with transparency and usability. This balance is particularly important in Human-Centered Information Retrieval systems, where user experience and interpretability are considered equally important as retrieval performance.

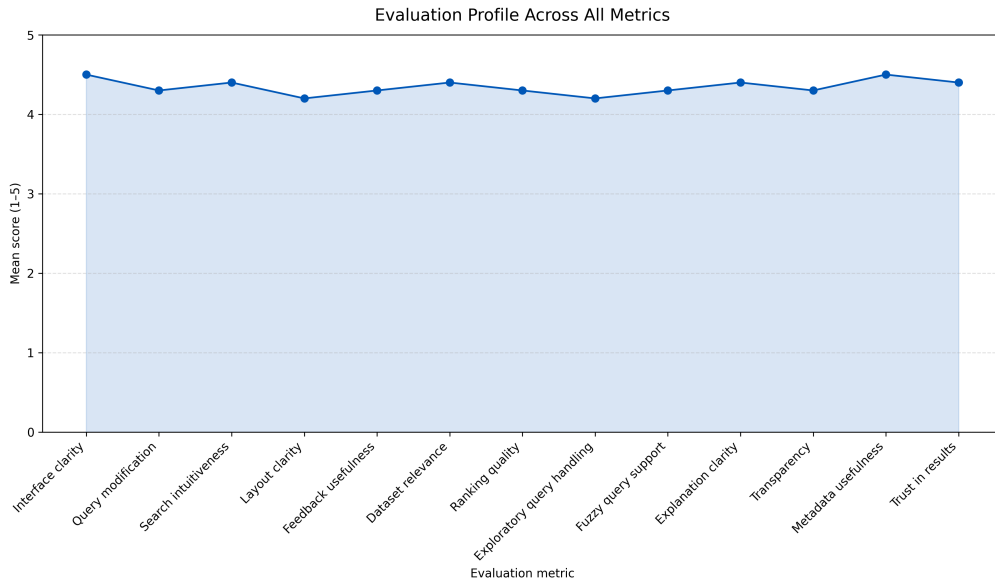


Figure 8: Evaluation profile of the proposed fuzzy HCIR framework across key assessment dimensions.

Figure 8 provides a consolidated view of the evaluation outcomes across the major assessment dimensions. The profile illustrates that all evaluated criteria received consistently positive scores, indicating strong overall acceptance of the prototype.

The balanced distribution of scores demonstrates that improvements in one dimension did not negatively affect other aspects of the system. For example, the addition of explainability features improved transparency without reducing usability, while the fuzzy ranking framework enhanced retrieval flexibility without introducing significant complexity.

6.6.1 Summary of Participant Feedback

Several representative comments collected during the evaluation illustrate the overall perception of the system:

- “The relevance scores made it easier to identify useful datasets quickly.”
- “The ranking explanations helped me understand why certain datasets appeared at the top of the results.”
- “The metadata presentation helped me evaluate datasets without opening multiple pages.”
- “The system still produced useful results even when my search query was vague or incomplete.”
- “The interface was easy to understand and required very little learning effort.”

Although the overall feedback was strongly positive, some participants suggested minor enhancements. These included additional filtering options for broad searches, improved differentiation between highly similar datasets, and short descriptions explaining how relevance scores are calculated. Importantly, no major usability or functionality concerns were identified during the evaluation.

Overall, the qualitative feedback supports the quantitative evaluation findings and further demonstrates the effectiveness of the proposed fuzzy HCIR framework. Participants consistently reported positive experiences regarding transparency, usability, relevance, and explainability, suggesting that the system successfully addresses several challenges commonly associated with exploratory Open Government Data retrieval.

7 Nielsen Heuristic Evaluation

The prototype was additionally evaluated using Nielsen’s usability heuristics, which provide a widely accepted framework for assessing interface usability, interaction quality, and user experience [9].

The heuristic assessment was conducted by the researcher as part of the Design Science Research evaluation process and was complemented by observations collected during participant testing sessions.

The objective of this evaluation was to determine whether the proposed fuzzy HCIR framework satisfies established usability principles while supporting exploratory dataset discovery and explainable retrieval.

The evaluation focused on five heuristics that were particularly relevant to the implemented retrieval system:

1. Visibility of System Status
2. Match Between System and the Real World
3. Consistency and Standards
4. Recognition Rather Than Recall
5. Transparency and User Trust

7.1 Visibility of System Status

The visibility of system status heuristic evaluates whether users receive clear and timely feedback regarding system operations and current states.

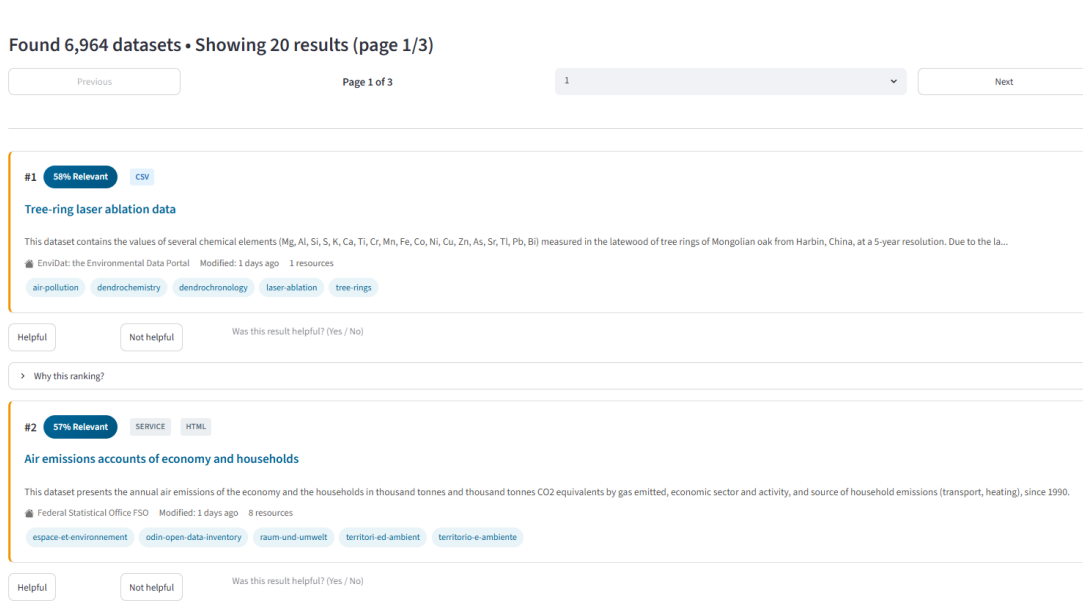


Figure 9: Example search interaction showing fuzzy query detection and retrieval feedback.

As illustrated in Figure 9, the prototype continuously provides feedback regarding query processing and retrieval status. The system informs users when fuzzy search behavior has been activated and clearly communicates the number of retrieved datasets.

Additional status indicators such as relevance scores, ranking positions, and meta-data summaries further improve visibility throughout the retrieval process. These features enable users to understand the current system state without requiring additional interaction.

Participants reported that the retrieval feedback messages and ranking indicators helped them understand how the system interpreted their queries and generated search results.

7.2 Match Between System and the Real World

The proposed retrieval framework was designed to communicate ranking decisions using concepts that are understandable to non-technical users.

Rather than presenting internal model parameters or algorithmic details, the system explains ranking outcomes through meaningful metadata attributes such as dataset completeness, publication recency, keyword similarity, and resource availability.

This design choice aligns the retrieval process with concepts that users naturally consider when evaluating datasets. As a result, participants were able to interpret ranking decisions without requiring knowledge of fuzzy logic or information retrieval algorithms.

The use of natural-language explanations contributed positively to transparency and improved users' understanding of retrieval behavior during exploratory search activities.

7.3 Consistency and Standards

Consistency is an important usability principle because it reduces cognitive effort and allows users to develop predictable interaction patterns.

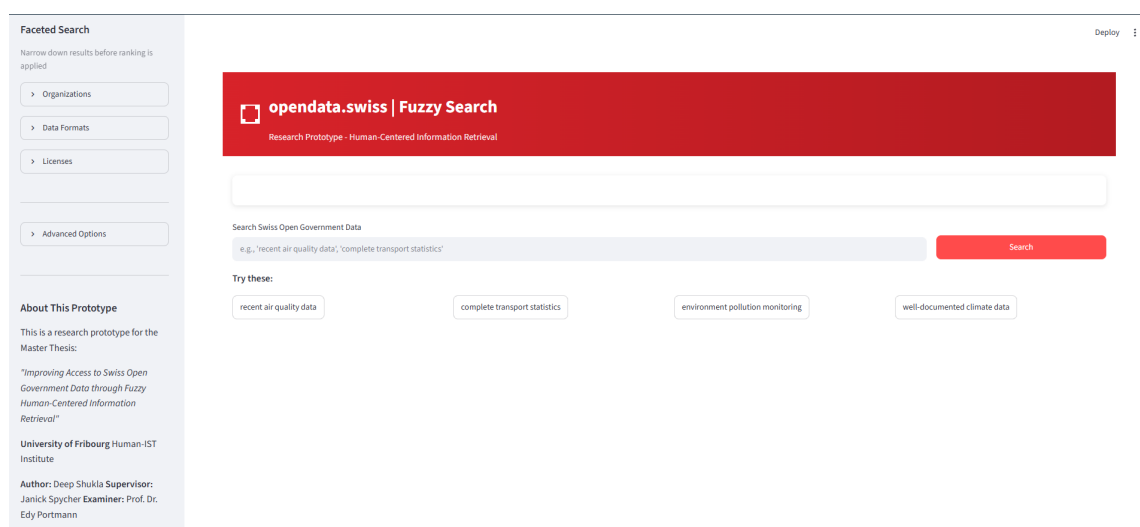


Figure 10: Consistent interface structure used throughout the retrieval process.

Figure 10 demonstrates the consistent layout employed throughout the retrieval system. Search controls, ranking indicators, metadata summaries, and explanation mechanisms are presented using a uniform visual structure.

Participants observed that result cards maintained consistent formatting across different search sessions. Relevance scores, metadata fields, tags, and ranking explanations were displayed using standardized presentation styles, which improved readability and reduced confusion.

The consistent interface design contributed positively to usability and supported efficient exploratory dataset discovery.

7.4 Recognition Rather Than Recall

The recognition rather than recall heuristic emphasizes minimizing memory requirements by making relevant information visible during interaction.

The proposed retrieval system supports this principle by presenting ranking scores, metadata summaries, tags, dataset descriptions, and explanation messages directly within the result interface.

Users are therefore not required to remember previous search results or metadata attributes when comparing datasets. Instead, important information remains visible throughout the search process, allowing participants to make informed decisions with minimal cognitive effort.

The evaluation results indicate that users were able to understand ranking behavior and dataset characteristics directly from the presented information without repeatedly revisiting previous search results.

7.5 Additional Evaluation Criterion: Transparency and User Trust

Although transparency and user trust are not part of Nielsen’s original ten usability heuristics, they were included as additional evaluation criteria because of their central importance in Human-Centered Information Retrieval and explainable information systems. The inclusion of these dimensions allowed the evaluation to assess whether the proposed retrieval framework successfully communicated ranking decisions and supported user confidence in the retrieved results.

Participants reported that the explanation mechanism contributed to higher perceived trust in the retrieval results. Participants reported that explicit ranking explanations helped them better understand ranking decisions and reduced uncertainty during exploratory search activities.

This observation aligns with previous research in explainable Artificial Intelligence and Human-Centered Information Retrieval, where transparent explanations are considered essential for improving interpretability and user trust [3, 8].

The heuristic evaluation demonstrates that the proposed fuzzy HCIR framework aligns well with established usability engineering principles. In particular, visibility of system status, consistency of interaction design, and metadata-driven explanations contributed positively to user experience and transparency.

The findings suggest that integrating explainable ranking mechanisms within exploratory retrieval systems can simultaneously improve usability, interpretability, and user trust. These observations further support the positive results obtained during the quantitative and qualitative evaluation phases.

8 Discussion

The evaluation results indicate that participants perceived the fuzzy HCIR prototype positively across usability, relevance, and explainability dimensions. Average ratings above 4.0 suggest that users found the interface intuitive and the ranking explanations useful for understanding search results.

The quantitative benchmark demonstrated that the proposed approach is competitive with traditional retrieval methods. Although the fuzzy ranking framework achieved a slightly lower aggregate nDCG score than the portal and BM25 baselines, the difference remained relatively small (0.60 compared with 0.66 and 0.63 respectively). The proposed framework therefore demonstrated competitive retrieval effectiveness while simultaneously providing additional support for exploratory search behaviour, metadata-aware ranking, and explainable retrieval.

The evaluation further highlighted the importance of transparency in dataset retrieval. Participants reported that ranking explanations helped them understand why datasets were retrieved and how different metadata factors influenced ranking decisions.

Together, these findings suggest that human-centered retrieval mechanisms can complement traditional keyword-based approaches by supporting exploratory search and improving user understanding of retrieval outcomes.

8.1 Comparison with Existing Literature

The findings of this evaluation are consistent with previous research in Human-Centered Information Retrieval and exploratory search systems [8]. Similar studies have demonstrated that interactive retrieval interfaces and transparent ranking explanations improve user engagement and trust.

The results also support prior work in explainable information retrieval and interpretable AI systems, where understandable ranking explanations contribute to improved interpretability and transparency [3].

Furthermore, the effectiveness of gradual multi-criteria relevance estimation observed in this evaluation aligns with previous fuzzy information retrieval research [7], where fuzzy reasoning improves retrieval flexibility for vague and exploratory queries.

8.2 Implications for Human-Centered Information Retrieval

The evaluation results demonstrate that combining fuzzy inference with explainable ranking mechanisms can significantly improve exploratory dataset retrieval in Open Government Data environments.

Unlike traditional keyword-based retrieval systems, the proposed HCIR framework supports gradual relevance estimation and more flexible interaction with vague or incomplete queries. This is particularly important in exploratory search scenarios, where users may not have clearly defined information needs at the beginning of the search process.

The integration of metadata-driven explanations also was associated with improved interpretability and transparency. Participants reported that explanation mechanisms helped them better understand ranking behavior and increased confidence in retrieval results. These findings support previous research emphasizing the importance of explainability and transparency in Human-Centered Information Retrieval systems [8, 3].

Furthermore, the fuzzy inference framework enabled the retrieval system to combine multiple metadata attributes simultaneously without relying solely on exact keyword matching. This improved the robustness of ranking behavior for exploratory search tasks and reduced the limitations commonly associated with strict Boolean retrieval models.

The evaluation additionally demonstrates that explainability and usability are closely connected in exploratory retrieval systems. Transparent ranking explanations not only improved interpretability but also supported user decision-making during dataset selection.

The discussion demonstrates that the proposed fuzzy HCIR framework successfully addresses several limitations commonly associated with traditional keyword-based retrieval systems. By combining fuzzy multi-criteria ranking, metadata-driven explanations, and user-centered interface design, the system provides a more interpretable and flexible retrieval experience.

The evaluation results further indicate that usability, explainability, and retrieval effectiveness are closely interconnected. Participants reported that improvements in transparency and explanation quality contributed to higher perceived trust and a more positive search experience. These findings reinforce the importance of Human-Centered Information Retrieval principles in the design of future Open Government Data discovery systems.

9 System Refinements After Evaluation

The evaluation phase generated valuable insights regarding system usability, transparency, and ranking behaviour. Following the Design Science Research methodology, these findings were used to refine the implemented prototype and improve overall artifact quality.

Most refinement activities focused on improving robustness, metadata handling, result presentation, and support for exploratory search behaviour. The objective was not to redesign the retrieval architecture but rather to address practical issues identified during user interaction.

Several participants reported situations where multilingual datasets contained incomplete title information. To address this issue, additional validation mechanisms were implemented to ensure that meaningful titles are always presented to users.

The evaluation also revealed occasional rendering issues caused by incomplete metadata fields. Additional filtering and normalization procedures were therefore introduced to improve result quality and reduce presentation inconsistencies.

Another refinement concerned exploratory search support. User feedback indicated that broad queries occasionally produced large result sets that were difficult to compare. Consequently, query interpretation and ranking procedures were adjusted to improve the prioritization of highly relevant datasets.

Improvements were also made to metadata presentation and result card design. These changes reduced visual clutter and improved readability without affecting the underlying retrieval process.

Overall, the implemented refinements improved usability, robustness, and transparency while preserving the original fuzzy retrieval architecture.

Issue Identified	Refinement Implemented
Invalid result rendering	Added validation filtering
Empty multilingual titles	Added safe title handling
Visual clutter in result cards	Simplified layout presentation
Broad exploratory queries	Improved query handling logic
Metadata inconsistencies	Added metadata normalization

Table 2: System refinements after evaluation

The refinements identified through evaluation were incorporated into the next iteration of the Design Science Research cycle. Feedback collected during prototype demonstrations, supervisor reviews, and user studies was used to refine ranking explanations, metadata visibility, query interpretation, and feedback mechanisms. These modifications informed the subsequent artifact iteration and contributed to the continuous improvement process prescribed by the DSR methodology.

Link to the Next DSR Iteration In accordance with Design Science Research principles, the findings obtained during the evaluation phase directly informed a subsequent cycle of artifact refinement. User feedback, usability observations, and retrieval evaluation results were incorporated into improvements of the ranking framework, explanation interface, and system interaction mechanisms. These refinements represent the next iteration of the Design Science Research process and contribute to the continuous evolution of the proposed retrieval framework.

10 Limitations

10.1 Evaluation Scale

The evaluation involved ten participants, which is appropriate for exploratory usability studies but limits the statistical generalizability of the findings. The relatively small sample size reduces statistical power and prevents the use of meaningful inferential statistical analyses. Consequently, the reported results should be interpreted as descriptive evidence regarding user perceptions rather than statistically generalizable conclusions.

Future evaluations involving larger participant populations would provide stronger empirical evidence and enable more robust statistical analysis of retrieval effectiveness, usability, and explainability outcomes.

10.2 Metadata-Centered Retrieval

The current framework primarily relies on metadata attributes rather than full semantic analysis of dataset contents. Although metadata-driven retrieval improves interpretability, deeper semantic representations could improve retrieval effectiveness for complex information needs.

10.3 Manual Fuzzy Rule Construction

The fuzzy inference system currently requires manual definition of linguistic variables and fuzzy rules. While this approach improves transparency and interpretability, automatic learning of fuzzy rules could improve scalability and adaptability in larger retrieval environments.

10.4 Heuristic Evaluation Procedure

The heuristic evaluation was not performed by multiple independent usability experts, as commonly recommended in Nielsen’s methodology. Instead, the assessment was conducted within the context of the design science research process and complemented by end-user feedback. Therefore, the findings should be interpreted as formative usability evidence rather than a formal expert heuristic evaluation.

10.5 External Validity

The participants involved in the user evaluation consisted primarily of Master’s-level students with backgrounds in computer science and information systems. Although this population provided valuable feedback regarding usability, relevance, and explainability,

it may not fully represent the broader population of Open Government Data users, such as journalists, policymakers, researchers, and public administrators.

Consequently, the findings should be interpreted with caution when generalising to other user groups. Future studies should involve more diverse participant populations to improve external validity.

Future evaluation work should also incorporate manually assigned graded relevance judgments produced by multiple independent assessors. Such an evaluation protocol would reduce potential bias introduced by automatic relevance estimation and provide a more reliable benchmark for comparing exploratory retrieval systems. In addition, inter-rater agreement measures could be reported to assess the consistency of relevance judgments across evaluators.

11 Research Contributions

This research contributes to the fields of Human-Centered Information Retrieval and Open Government Data discovery through the design, implementation, and evaluation of a novel fuzzy retrieval framework.

First, the research proposes a fuzzy HCIR architecture capable of supporting exploratory dataset retrieval through gradual multi-criteria relevance estimation.

Second, the work demonstrates how metadata-driven explainability mechanisms can improve transparency, interpretability, and user trust during dataset discovery.

Third, a fully functional prototype was implemented and evaluated using real datasets obtained from the Swiss Open Government Data portal.

Fourth, the research provides empirical evidence that fuzzy reasoning can improve the handling of vague and exploratory search queries while maintaining high levels of usability and explainability.

Finally, the work illustrates how Design Science Research principles can be applied to iteratively design, evaluate, and refine retrieval systems intended for exploratory information seeking.

12 Conclusion

This thesis presented and evaluated a fuzzy Human-Centered Information Retrieval framework for exploratory access to Swiss Open Government Data.

The proposed system combines metadata-driven retrieval, fuzzy multi-criteria ranking, and explainable ranking mechanisms within an interactive search environment. A functional prototype was implemented and evaluated using quantitative benchmarking, user-centered evaluation, and heuristic assessment.

The benchmark evaluation demonstrated that the proposed fuzzy HCIR framework achieved retrieval effectiveness that was competitive with both the portal ranking and BM25 baselines while providing additional benefits in terms of exploratory search support, metadata-aware ranking, transparency, and explainability. User evaluation results further indicated positive perceptions of usability, relevance, and explainability.

The Design Science Research methodology enabled iterative refinement of the artifact through multiple evaluation cycles. Feedback obtained during prototype demonstrations, supervisory reviews, and user studies informed improvements to ranking transparency, metadata presentation, and query interpretation.

Future work should incorporate manually generated relevance judgments, larger and more diverse participant populations, and hybrid retrieval techniques that combine fuzzy reasoning with semantic search approaches.

Overall, the evaluation demonstrated that the proposed fuzzy Human-Centered Information Retrieval framework provides retrieval performance that is competitive with traditional ranking approaches while offering additional benefits in terms of explainability, transparency, and support for exploratory search behaviour. Although the benchmark evaluation revealed limitations in the current measurement methodology, the results indicate that fuzzy multi-criteria retrieval represents a promising direction for Open Government Data discovery and warrants further investigation using larger-scale and human-judged evaluation protocols.

A Evaluation Questionnaire

The evaluation questionnaire used during the user-centered evaluation consisted of usability, relevance, explainability, and satisfaction-related questions based on a 5-point Likert scale.

The following scale was used:

- 1 = Strongly disagree
- 2 = Disagree
- 3 = Neutral
- 4 = Agree
- 5 = Strongly agree

Example questionnaire items included:

- The interface is easy to understand.
- The search process felt intuitive.
- The returned datasets were relevant to my search queries.
- The explanations for ranking decisions were understandable.
- The relevance scores improved transparency and trust.

B Example Search Queries

The following exploratory queries were used during evaluation:

- transport datasets
- climate statistics
- healthcare data

- environmental monitoring
- population statistics
- public mobility data

Participants were also encouraged to perform free exploratory searches using topics of personal interest.

C Example Generated Explanations

The prototype generated metadata-driven explanations describing why datasets were ranked highly.

Example explanation:

“This dataset was ranked highly because it closely matched the query terms, contained complete metadata, and was recently updated.”

Another example explanation:

“The dataset received a high relevance score because of strong metadata completeness, keyword similarity, and recent publication activity.”

D Example User Feedback

The following representative comments were collected from participants during the evaluation process:

- “The relevance scores made it easier to identify the most useful datasets quickly.”
- “I liked how the system explained why certain datasets appeared at the top of the ranking.”
- “The metadata presentation helped me decide whether a dataset was relevant without opening multiple pages.”
- “The interface was straightforward and easy to understand, even during my first use.”
- “The system performed well when I used broad or exploratory search terms.”
- “The explanations increased my confidence in the search results and made the ranking process more transparent.”
- “The search experience felt smooth and the results were generally well organized.”
- “Additional filtering options could further improve searches that return many similar datasets.”
- “A short explanation of how relevance scores are calculated could improve transparency for first-time users.”
- “For some broad searches, many datasets appeared relevant, but the ranking still helped identify the most useful options.”

References

- [1] J. Attard, F. Orlandi, S. Scerri, and S. Auer, “A systematic review of open government data initiatives,” *Government Information Quarterly*, vol. 32, no. 4, pp. 399–418, 2015. [Online]. Available: <https://doi.org/10.1016/j.giq.2015.07.006>
- [2] J. Brooke, “Sus: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland, Eds. Taylor Francis, 1996, pp. 189–194.
- [3] F. Doshi-Velez and B. Kim, “Towards a rigorous science of interpretable machine learning,” *arXiv preprint arXiv:1702.08608*, 2017. [Online]. Available: <https://doi.org/10.48550/arXiv.1702.08608>
- [4] A. R. Hevner, S. T. March, J. Park, and S. Ram, “Design science in information systems research,” *MIS Quarterly*, vol. 28, no. 1, pp. 75–105, 2004. [Online]. Available: <https://doi.org/10.2307/25148625>
- [5] M. Janssen, Y. Charalabidis, and A. Zuiderwijk, “Benefits, adoption barriers and myths of open data and open government,” *Information Systems Management*, vol. 29, no. 4, pp. 258–268, 2012. [Online]. Available: <https://doi.org/10.1080/10580530.2012.716740>
- [6] K. Järvelin and J. Kekkonen, “Cumulated gain-based evaluation of ir techniques,” *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002. [Online]. Available: <https://doi.org/10.1145/582415.582418>
- [7] D. H. Kraft and E. Colvin, *Fuzzy Information Retrieval*. Springer, 2017.
- [8] G. Marchionini, “Exploratory search: From finding to understanding,” *Communications of the ACM*, vol. 49, no. 4, pp. 41–46, 2006. [Online]. Available: <https://doi.org/10.1145/1121949.1121979>
- [9] J. Nielsen, *Usability Engineering*. Morgan Kaufmann, 1994.
- [10] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: Bm25 and beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009. [Online]. Available: <https://doi.org/10.1561/15000000019>
- [11] L. A. Zadeh, “Fuzzy sets,” *Information and Control*, vol. 8, no. 3, pp. 338–353, 1965. [Online]. Available: [https://doi.org/10.1016/S0019-9958\(65\)90241-X](https://doi.org/10.1016/S0019-9958(65)90241-X)